



Data Management and Visualization

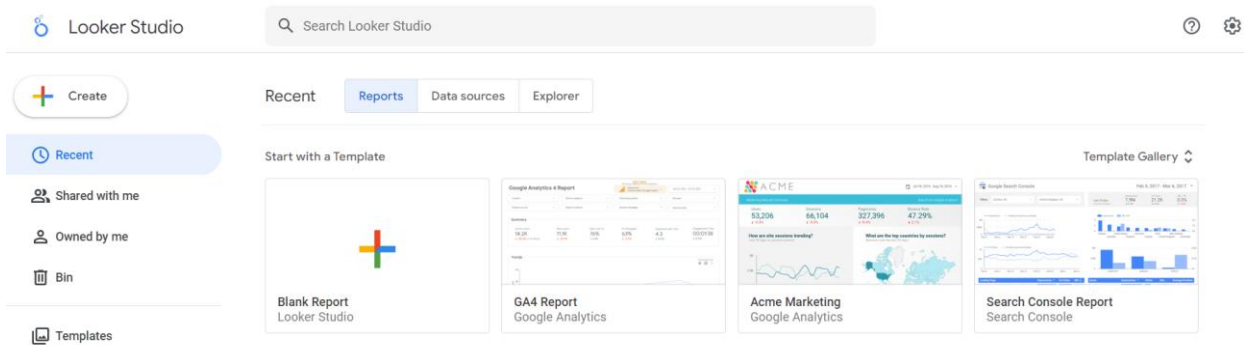
Politecnico di Torino

Looker Studio – Practice 3

1. Login

Connect to [Google Looker Studio](#), login with your Google Account or [create a new free Google Account](#).

<https://lookerstudio.google.com>



2. Welcome report

Click on “Tutorial Report” and follow the tutorial.

- https://support.google.com/looker-studio/topic/6289358?hl=en&ref_topic=9170843&sjid=16433828157045416046-EU
- learn the basics of the Looker Studio tool by copying the “Welcome report” and following the step-by-step instructions provided.



3. New report on Airbnb Boston reviews

To create a new report from scratch, a data source must be identified. To this aim, a portion of the [Kaggle dataset of the Airbnb reviews in Boston](#) has been uploaded into a [shared Google Sheets](#) to be used as data source for Google Looker Studio.

- the Google Sheets, with approximately 10k reviews to be used as data source, is available at <https://docs.google.com/spreadsheets/d/1a2c9vCMFFfDXmhjoEoX2EwS2IYTbqE4WfZY72TXW9co/edit#gid=285360760>

AirBnB reviews in Boston by Kaggle

File Edit View Insert Format Data Tools Add-ons Help All changes saved in Drive

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listing_id

	A	B	C	D	E	F	G
	listing_id	id	Date	reviewer_id	reviewer_name	comments	host_id
1							
2	9857	92488772	08-Aug-16	56470796	Danielle	Seamless was easy to work with and very responsive. The apt was clean, spacious, and in walking distance to dining, shopping, and nightlife. We would stay here again.	25188
3	9857	90834004	02-Aug-16	86373945	Courtney	Seamless was really great about communication with my family. The location is absolutely perfect and close to everything! It's a little noisy from the road, and the air conditioner kept fluctuating. The bathroom shower curtain looked like maybe mold was growing on it. But other than that it was a good place to stay in!	25188
4	9857	81205609	21-Jun-16	74520248	Patrick	We had a fantastic stay in Boston. Apartment presented exactly like the photos and the location is just perfect for exploring Boston. It was straight forward to check in and I would not hesitate to stay again if I was to return to Boston.	25188
5	9857	52407219	29-Oct-15	42949625	Heather	Excellent location, good sized two bedroom apartment, kitchen had everything you need. Overall would definitely stay here again!	25188
6	9857	51958656	25-Oct-15	45617313	John	The apartment was even better than the pictures, very comfortable and nice, check-in was very easy and the apartment was clean and had plenty of amenities (bathroom supplies, towels, soap). I would recommend this location and apartment to anyone who is looking for a place to stay in Boston.	25188
7	9857	46422364	10-Sep-15	2642405	Mike	Awesome place, nicer than the pictures, super clean and the AC was on when we got it (was 90 outside!)	25188
8	9857	37160926	05-Jul-15	11002414	Stephanie	The name seamless says it all. The whole process couldn't have been easier or more pleasant. The apartment was in an amazing location and was clean and very comfortable to stay in. There are two wall units for AC which was completely sufficient to cool the entire apartment. We were extremely happy with our experience and will definitely consider staying there again during our next trip to Boston.	25188
9	9857	33730553	30-Mar-15	15030304	Kristina	Everything worked perfect, from checkin to checkout. The apartment was clean and comfortable.	25188

Reviews Query DW Listings Sheet1

- Spend some time to understand the data by reading their description on Kaggle and looking at the table on Google Sheets.
- The data source table has been created by joining the “Listings” and “Reviews” original tables provided by Kaggle and exporting the first 10k joined rows sorted by ascending “listing_id”.

Data sources

Data sources have two types of fields: dimensions and metrics.

- A **dimension** is a categorical data.
- A **metric** is a number that quantifies something in that category.
- A Looker Studio report lets you visualize those dimensions and metrics in charts and tables.
- In your Looker Studio data sources and report properties panels, dimensions appear as **green** chips, while metrics appear as **blue** chips.

Field	ID
Browser	t0_ga:browser_
Sessions	t0_ga:sessions_

Create a new report

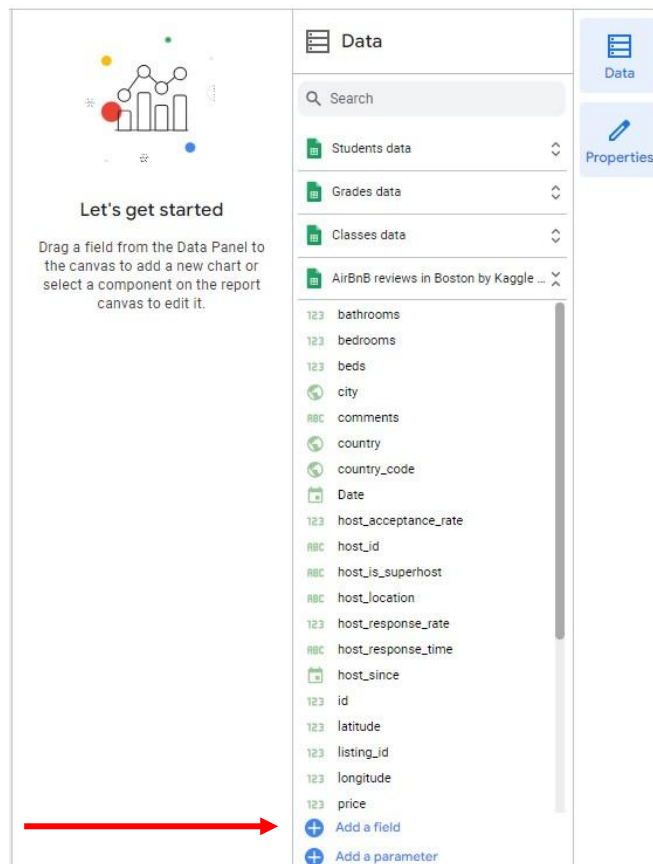
- Go to the Looker Studio home page.
- Click on “Start a new report” (Blank).
- Rename the “Untitled Report” with a name of your choice by clicking on the name itself.



- Create a new data source by clicking on the button on the top left or select the Airbnb data source if it is already present in the right-pane list.
- Connect to the Google Sheet data source by using its URL:
 - o Choose the “Google Sheets” connector from the list of possible connectors
 - o Choose the “URL” option in the first column
 - o Paste the Airbnb-data Google Sheet URL in the specific field:
<https://docs.google.com/spreadsheets/d/1a2c9vCMFFfDXmhjoEoX2EwS2lYTbqE4WfZY72TXW9co/edit?usp=sharing>
 - o Choose the “Reviews Query DW” worksheet in the next column
 - o Tick the option to “use the first row as headers” if it is not ticked yet
 - o Click on the “Add” button to execute the connection to the data source

Dimensions, metrics, and transformations

- Check the **type** and **aggregation** of each field. Measures and dimensions are interpreted **automatically**. Although, you can safely go on if you think that they have been misinterpreted.
- **Create new useful fields (dimensions or metrics)** from the existing ones by exploiting formulas, such as in the following (click on the “+” and “fx” placeholders). For details on this step, see: <https://support.google.com/datastudio/answer/6299685?hl=en>



- **LENGTH**(comments) → to count the number of chars of the comment field
- **CONCAT**(latitude, CONCAT(' ', longitude)) → to generate a (lat, long) field useful for map charts; before generating this new field, set “**Type=Text**” for latitude and longitude fields, so that they become dimensions (by default, Looker Studio considers them as metrics)
- **price / square_feet** → to compute the average price per square feet (try to create a field that contains the square meters instead of the square feet (1 foot = 0.3048 meter)).
- **MONTH**(Date) → to extract the month of the year from the full date, e.g. 12
- **YEAR**(Date) → to extract the year from the full date, e.g. 2017
- **CONCAT**(YEAR(Date), MONTH(Date)) → to build a field which is the full month, e.g. 201712
 - if you already have the computed fields “month” and “year”, you can also use them in the formula, e.g., CONCAT(year, month)

← EDIT CONNECTION						
Index	Field		Type		Aggregation	
21	property_type	::	ABC Text	▼	None	
22	room_type	::	ABC Text	▼	None	
23	bathrooms	::	123 Number	▼	None	▼
24	bedrooms	::	123 Number	▼	None	▼
25	beds	::	123 Number	▼	None	▼
26	square_feet	::	123 Number	▼	None	▼
27	price	::	123 Number	▼	None	▼
28	review_scores_rating	::	123 Number	▼	None	▼
29	review_scores_value	::	123 Number	▼	None	▼
30	comment_length	fx ::	123 Number	▼	None	▼
31	latlong	fx ::	🌐 Latitude, Longitude	▼	None	
32	price_per_ft2	fx ::	123 Number	▼	None	▼
33	month	fx ::	📅 Month (MM)	▼	None	
34	year	fx ::	📅 Year (YYYY)	▼	None	
35	month_year	fx ::	ABC Text	▼	None	

↻ REFRESH FIELDS

After creating new fields and updating the existing ones, click on “Add to report”

Analyse the data

Analyse the data by building the following visualizations. Then, explore and create new visualizations to find interesting insights on your own.

- **Analysis (1):** compare the trend of the average length of the review “comments” (number of chars) vs the average “review_scores_rating” for different “property_type”. Sort the data by descending average length of comments. Allow end-users to filter the data under analysis by selecting a date range of their choice.



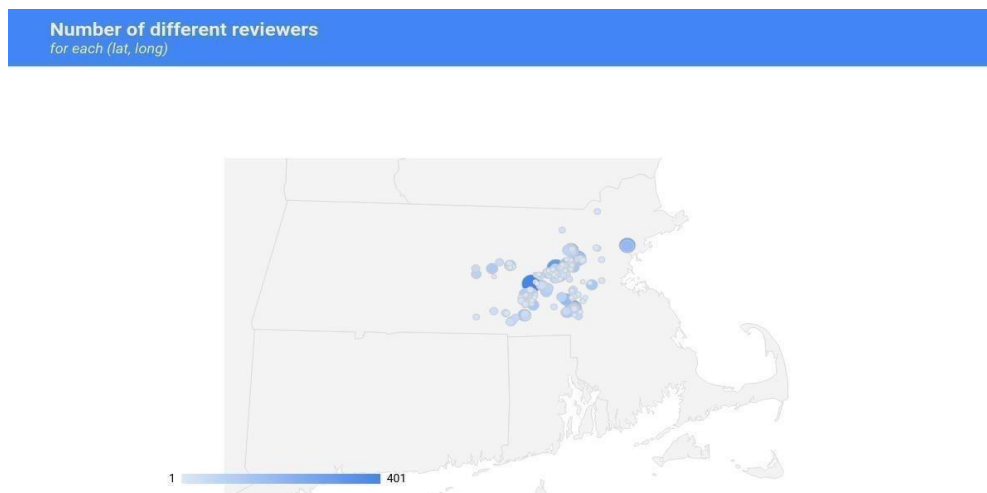
- **Analysis (2):** compare the trend of the number of different “listing_id” reviewed, for each “room_type”, and for each month of the year. Allow end-users to filter the data under analysis by selecting a date range and the type of superhost (true/false).



Explore, create, and present new additional analyses to identify interesting insights. For instance:

- **Analysis (3):** analyse the number of different reviewers for each location (lat, long).

Note that the Kaggle dataset of the Airbnb reviews is in Boston, **Massachusetts**, US



- **Analysis (4):** Visualize, for each property type and for each year, the average rating score values sorted by ascending property type and by descending mean rating_score_value. Exclude possible null values for the attribute property_type.

Average review score values for each property type and year			
	property_type ⓘ ▲	year	review_scores_value ⓘ ▼
1.	Apartment	2009	9.38
2.	Apartment	2010	9.04
3.	Apartment	2016	9.03
4.	Apartment	2014	9
5.	Apartment	2013	8.99
6.	Apartment	2015	8.97
7.	Apartment	2011	8.94
8.	Apartment	2012	8.92
9.	Bed & Breakfast	2014	8.94
10.	Bed & Breakfast	2013	8.93

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- **Analysis (5):** Visualize, for each year and for each room type, the total count of top-scored reviews (review_score_value = 10).

Compare the obtained results with the count of the distinct listing_id reviewed.

